

Semester	B.E. Semester VII – CMPN
Subject	R Programming LAB
Subject Professor In-charge	Prof. Sanjeev Dwivedi
Assisting Teachers	
Laboratory	M411

Student Name	Eshanika Amballa
Roll Number	23102B0056
Grade and Subject Teacher's Signature	

Experiment Number	07
Experiment Title	Customer Segmentation and Predictive Analytics Using Machine Learning
Program	<pre> pkgs <- c("tidyverse", "readxl", "factoextra", "cluster", "caret", "randomForest", "e1071", "pROC", "plotly", "htmlwidgets") new <- pkgs[!pkgs %in% installed.packages()[, "Package"]] if (length(new)) install.packages(new) invisible(lapply(pkgs, library, character.only = TRUE)) set.seed(42) if (!file.exists("Online Retail.xlsx")) { download.file("https://archive.ics.uci.edu/static/public/352/online+ retail.zip", "online_retail.zip", mode = "wb") unzip("online_retail.zip") } raw <- read_excel("Online Retail.xlsx") glimpse(raw) cat("Rows:", nrow(raw), "\n") print(colSums(is.na(raw))) # CustomerID has ~135k missing # 2. Cleaning </pre>

```

clean <- raw %>%
  mutate(InvoiceNo = as.character(InvoiceNo),
         Date = as.Date(InvoiceDate)) %>%
  filter(!is.na(CustomerID),
         !str_starts(InvoiceNo, "C"),
         Quantity > 0, UnitPrice > 0) %>%
  mutate(Revenue = Quantity * UnitPrice)

cat("Rows after cleaning:", nrow(clean), "\n")
cat("Customers:", n_distinct(clean$CustomerID), "\n")

# 3. Customer-level feature engineering (RFM + extras)
snapshot <- max(clean$Date) + 1

cust <- clean %>%
  group_by(CustomerID) %>%
  summarise(
    Recency      = as.numeric(snapshot - max(Date)),
    Frequency    = n_distinct(InvoiceNo),
    Monetary     = sum(Revenue),
    TotalQuantity = sum(Quantity),
    TenureDays   = as.numeric(max(Date) - min(Date)) + 1,
    .groups = "drop"
  ) %>%
  mutate(
    AvgTransactionValue = Monetary / Frequency,
    # invoices per month; tenure floored at 1 month so one-time
    buyers
    # don't get an inflated rate
    PurchaseFrequency = Frequency / pmax(TenureDays / 30,
    1)
  )

feat_cols <- c("Recency", "Frequency", "Monetary",
              "AvgTransactionValue",
              "TotalQuantity", "PurchaseFrequency")
summary(cust[, feat_cols])

# 4. Outlier treatment + scaling
# Features are heavily right-skewed: log1p first, then cap at
# 1.5*IQR
cap_iqr <- function(x) {
  q <- quantile(x, c(.25, .75)); i <- q[2] - q[1]

```

```

    pmin(pmax(x, q[1] - 1.5 * i), q[2] + 1.5 * i)
  }
  X_log <- cust %>% select(all_of(feats_cols)) %>%
    mutate(across(everything(), ~ cap_iqr(log1p(.x))))
  X <- as.data.frame(scale(X_log))      # z-score standardisation

  par(mfrow = c(1, 2))
  boxplot(cust$Monetary, main = "Monetary (raw)")
  boxplot(X_log$Monetary, main = "Monetary (log + capped)")
  par(mfrow = c(1, 1))

  # 5. Elbow method + silhouette sweep
  d <- dist(X)                      # reused for silhouettes / hclust
  print(fviz_nbclust(X, kmeans, method = "wss", k.max = 10, nstart
    = 25) +
    labs(title = "Elbow Method"))
  print(fviz_nbclust(X, kmeans, method = "silhouette", k.max = 10,
    nstart = 25) +
    labs(title = "Average Silhouette by k"))

  # 6. K-Means
  K <- 4      # <- set from the elbow / silhouette plots
  km <- kmeans(X, centers = K, nstart = 50)
  cust$Cluster_KM <- km$cluster
  table(cust$Cluster_KM)

  # 7. Hierarchical clustering + dendrogram
  hc <- hclust(d, method = "ward.D2")
  plot(hc, labels = FALSE, hang = -1, main = "Dendrogram
    (Ward.D2)")
  rect.hclust(hc, k = K, border = 2:(K + 1))
  cust$Cluster_HC <- cutree(hc, k = K)

  # 8. Silhouette comparison + agreement
  sil_km <- silhouette(cust$Cluster_KM, d)
  sil_hc <- silhouette(cust$Cluster_HC, d)
  sil_tbl <- data.frame(Method = c("K-Means", "Hierarchical
    (Ward)"),
    AvgSilhouette = c(mean(sil_km[, 3]), mean(sil_hc[,
    3])))
  print(sil_tbl)
  print(fviz_silhouette(sil_km))

  ari <- function(a, b) {          # Adjusted Rand Index

```

```

tab <- table(a, b); n <- sum(tab); c2 <- function(x) x * (x - 1) / 2
s <- sum(c2(tab)); sa <- sum(c2(rowSums(tab))); sb <-
sum(c2(colSums(tab)))
e <- sa * sb / c2(n); (s - e) / ((sa + sb) / 2 - e)
}
print(table(KMeans = cust$Cluster_KM, Hierarchical =
cust$Cluster_HC))
cat("Adjusted Rand Index (KM vs HC):",
round(ari(cust$Cluster_KM, cust$Cluster_HC), 3), "\n")

# 9. PCA + 2D visualisation
pca <- prcomp(X, center = FALSE, scale. = FALSE) # X already
scaled
print(summary(pca))
print(fviz_eig(pca, addlabels = TRUE))
print(fviz_pca_var(pca, repel = TRUE)) # loadings

scores <- as.data.frame(pca$x[, 1:3])
scores$Cluster <- factor(cust$Cluster_KM)
ggplot(scores, aes(PC1, PC2, colour = Cluster)) +
  geom_point(alpha = .6, size = 1.4) +
  stat_ellipse(level = .9) +
  labs(title = "Customer Segments - 2D PCA Projection") +
  theme_minimal()

# 10. Cluster profiling
profile <- cust %>%
  group_by(Cluster_KM) %>%
  summarise(Customers = n(),
            Pct = round(100 * n() / nrow(cust), 1),
            across(all_of(feats_cols), median), # median: robust to
            outliers
            TotalRevenue = sum(Monetary),
            .groups = "drop") %>%
  mutate(RevenueShare = round(100 * TotalRevenue /
sum(TotalRevenue), 1))
print(as.data.frame(profile))
# Use this table to NAME your clusters (e.g. Champions, Loyal,
At-Risk,
# Lost/One-time). Naming is a judgement call: write it in your
notebook.

# 11. High-value target
hv_cluster <- profile$Cluster_KM[which.max(profile$Monetary)]

```

```
cat("High-value cluster:", hv_cluster, "\n")

model_df <- X
model_df$Target <- factor(ifelse(cust$Cluster_KM == hv_cluster,
                                "HighValue", "Other"),
                          levels = c("Other", "HighValue"))
print(table(model_df$Target))

idx <- createDataPartition(model_df$Target, p = 0.7, list =
FALSE)
train <- model_df[idx, ]
test <- model_df[-idx, ]

# 12. Models: Random Forest + SVM
rf <- randomForest(Target ~ ., data = train, ntree = 500,
importance = TRUE)
rf_pred <- predict(rf, test)
rf_prob <- predict(rf, test, type = "prob")[, "HighValue"]

tuned <- tune.svm(Target ~ ., data = train, kernel = "radial",
                  cost = c(0.5, 1, 5, 10), gamma = c(0.05, 0.1, 0.5),
                  tunecontrol = tune.control(cross = 5))
svm_fit <- svm(Target ~ ., data = train, kernel = "radial",
probability = TRUE,
              cost = tuned$best.parameters$cost,
              gamma = tuned$best.parameters$gamma)
svm_raw <- predict(svm_fit, test, probability = TRUE)
svm_pred <- svm_raw
svm_prob <- attr(svm_raw, "probabilities")[, "HighValue"]

# 13. Evaluation
evaluate <- function(name, pred, prob, truth) {
  cm <- confusionMatrix(pred, truth, positive = "HighValue")
  rc <- roc(truth, prob, levels = c("Other", "HighValue"), direction
= "<", quiet = TRUE)
  row <- data.frame(Model = name,
                    Accuracy = unname(cm$overall["Accuracy"]),
                    Precision = unname(cm$byClass["Precision"]),
                    Recall = unname(cm$byClass["Recall"]),
                    F1 = unname(cm$byClass["F1"]),
                    ROC_AUC = as.numeric(auc(rc)))
  list(cm = cm, roc = rc, row = row)
}
```

```

ev_rf <- evaluate("Random Forest", rf_pred, rf_prob,
test$Target)
ev_svm <- evaluate("SVM (RBF)", svm_pred, svm_prob,
test$Target)

results <- rbind(ev_rf$row, ev_svm$row)
print(results, digits = 3)
print(ev_rf$cm$table); print(ev_svm$cm$table) # confusion
matrices

plot(ev_rf$roc, col = "forestgreen", main = "ROC Curves")
plot(ev_svm$roc, col = "firebrick", add = TRUE)
legend("bottomright", c("Random Forest", "SVM"), col =
c("forestgreen", "firebrick"), lwd = 2)

# 14. Feature importance (Random Forest)
imp <- as.data.frame(importance(rf))
imp$Feature <- rownames(imp)
ggplot(imp, aes(reorder(Feature, MeanDecreaseGini),
MeanDecreaseGini)) +
  geom_col(fill = "steelblue") + coord_flip() +
  labs(title = "Random Forest Feature Importance", x = NULL) +
  theme_minimal()

# Sanity check for label leakage: the target comes from clusters
built on
# these same features, so near-perfect scores are expected.
Retrain without
# the most direct value features to see how much behaviour
alone predicts.
lite_cols <- setdiff(feats_cols, c("Monetary",
"AvgTransactionValue", "TotalQuantity"))
rf_lite <- randomForest(Target ~ ., data = train[, c(lite_cols,
"Target")], ntree = 500)
ev_lite <- evaluate("RF (no Monetary/AvgTxn/Qty)",
  predict(rf_lite, test),
  predict(rf_lite, test, type = "prob")[, "HighValue"],
test$Target)
print(ev_lite$row, digits = 3)

# 15. Interactive 3D (Plotly)
fig <- plot_ly(scores, x = ~PC1, y = ~PC2, z = ~PC3, color =
~Cluster,
  type = "scatter3d", mode = "markers",

```

	<pre>marker = list(size = 3, opacity = 0.7)) %>% layout(title = "Customer Clusters - 3D PCA") fig saveWidget(fig, "clusters_3d.html", selfcontained = TRUE)</pre>																				
Output	• Clustering quality: K-Means has a silhouette of 0.267 and hierarchical 0.191. That is modest, meaning the segments overlap and RFM data is more of a continuum than clean blobs. Say that honestly. K-Means fits better, and an ARI of 0.50 means the two methods agree only moderately. • PCA: the first three components explain about 90% of the variance, so the 2D and 3D plots are a faithful view. PC1 alone is 56%. • Profiles: <table><tr><th>Cluster</th><th>Size</th><th>Median profile</th><th>Suggested name</th></tr><tr><td>4</td><td>18%</td><td>10 days since last order, 9 orders, ~£3,700</td><td>Champions (the</td></tr><tr><td>1</td><td>30%</td><td>36 days, 3 orders, ~£1,020</td><td>Developing regu</td></tr><tr><td>3</td><td>27%</td><td>86 days, 1 order, ~£526, large baskets</td><td>One-time big sp</td></tr><tr><td>2</td><td>24%</td><td>152 days, 1 order, ~£174</td><td>Lapsed low-valu</td></tr></table> • Models: SVM scores 0.997 accuracy and 1.000 AUC. Random Forest scores 0.982 accuracy and 0.923 recall, because it misses 18 of the high-value customers. SVM is better here, but both scores are inflated, because the label came from clustering these same features. The reduced Random Forest (without Monetary, AvgTransactionValue and TotalQuantity) still gets AUC 0.975 and F1 0.84, so behaviour alone predicts the segment well. This is your critical-reasoning paragraph. • Feature importance: show the plot. If Monetary, Frequency and Recency lead, that matches how the segments were defined.	Cluster	Size	Median profile	Suggested name	4	18%	10 days since last order, 9 orders, ~£3,700	Champions (the	1	30%	36 days, 3 orders, ~£1,020	Developing regu	3	27%	86 days, 1 order, ~£526, large baskets	One-time big sp	2	24%	152 days, 1 order, ~£174	Lapsed low-valu
Cluster	Size	Median profile	Suggested name																		
4	18%	10 days since last order, 9 orders, ~£3,700	Champions (the																		
1	30%	36 days, 3 orders, ~£1,020	Developing regu																		
3	27%	86 days, 1 order, ~£526, large baskets	One-time big sp																		
2	24%	152 days, 1 order, ~£174	Lapsed low-valu																		







